**1.** Given the line $\vec{r} = [12,-8,-4] + t[-3,4,2]$, find the intersection(s) with the xy plane.

**2.** Find vector and parametric equations of the plane that contains the two intersecting lines ,

$\vec{r} = [3,-1,2] + s[4,0,1]$ and , $\vec{r} = [3,-1,2] + t[4,0,2]$.

**3.** Find a scalar equation of the plane that contains the origin and the point (2, -3, 2) and is perpendicular to the plane $x + 2y - z + 3 = 0$.

**4.** Determine the parametric equations for the plane containing the 2 parallel lines:
$L_1 : \vec{r} = [0,1,3] + t[-6,-3,6]$ and $L_2 : \vec{r} = [-4,5,-4] + s[4,2,-4]$.

**5.** Determine the **scalar** equation of the plane parallel to the plane $\pi_1 : 3x + y - 2z - 4 = 0$ and containing the point of intersection of lines $L_1 : x = 7 + 2s, y = 2 + s, z = -6 - 3s$ and $L_2 : \vec{r} = [3,9,13] + t[1,5,5]$.

**6.** Find the scalar equation of the plane that is perpendicular to the plane $\vec{r} = [4,-5,2] + s[2,1,3] + t[-1,4,0]$ and intersects it at the line $\vec{r} = [4,-5,2] + t[1,-1,1]$.

**7.** Consider the lines $l_1 : \vec{r} = [1,-2,4] + s[1,1,-3]$ and $l_2 : \vec{r} = [4,-2,k] + t[2,3,1]$ Determine the equation of the plane that contains $l_1$ and is parallel to $l_2$ .

**8.** Find the equation of the plane that contains the line $\vec{r} = [3,1,0] + t[2,1,4]$ and is perpendicular to the plane $\pi : \vec{p} = [1,1,1] + k[1,0,5] + s[-4,2,3]$.

**9.** Determine the measure of the obtuse angle to the nearest degree between the lines:
$l_1 : \vec{r} = [1,-2,3] + t[4,1,-1]$ and $l_2 : \ x = 3 - t \ , \ y = t \ , z = 3 + 2t$.

**10.** Determine the Cartesian equation of the plane that contains the point (2,-1,1) and is perpendicular to the line joining points (-1,3,2) and (4,0,-2).

**11.** Determine the value of k if the acute angle between the planes 2x+ky-8=0 and x-3y+z+4=0 is $60^\circ$.